

ΕΘΝΙΚΟ ΜΕΤΣΟΒΙΟ ΠΟΛΥΤΕΧΝΕΙΟ

ΣΧΟΛΗ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ

ΕΡΓΑΣΤΗΡΙΟ ΗΛΕΚΤΡΟΝΙΚΩΝ ΑΙΣΘΗΤΗΡΙΩΝ

Εργαστηριακή Άσκηση

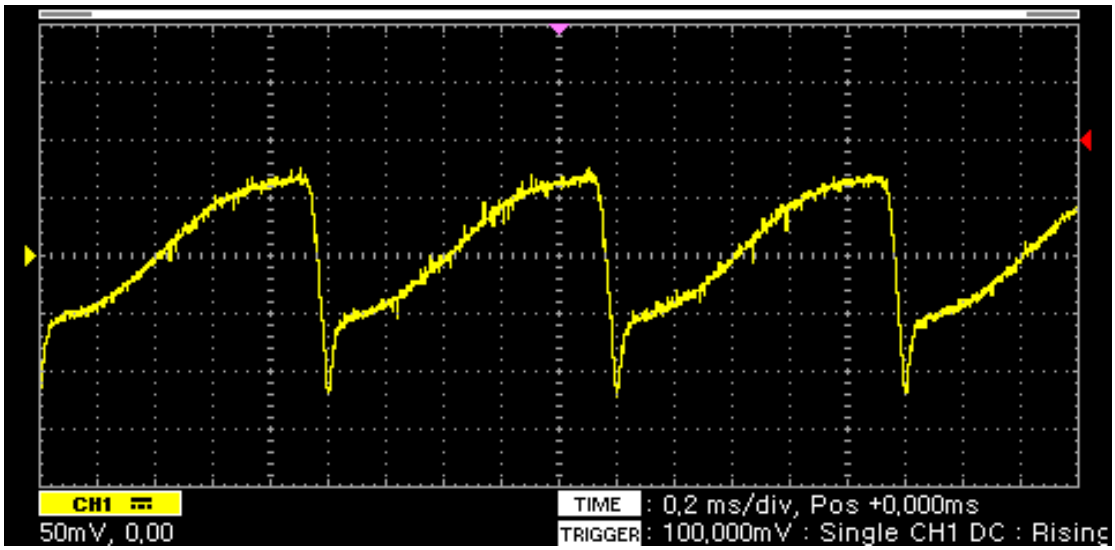
Βρόχος υστέρησης άμορφων υλικών

Αθήνα
2022

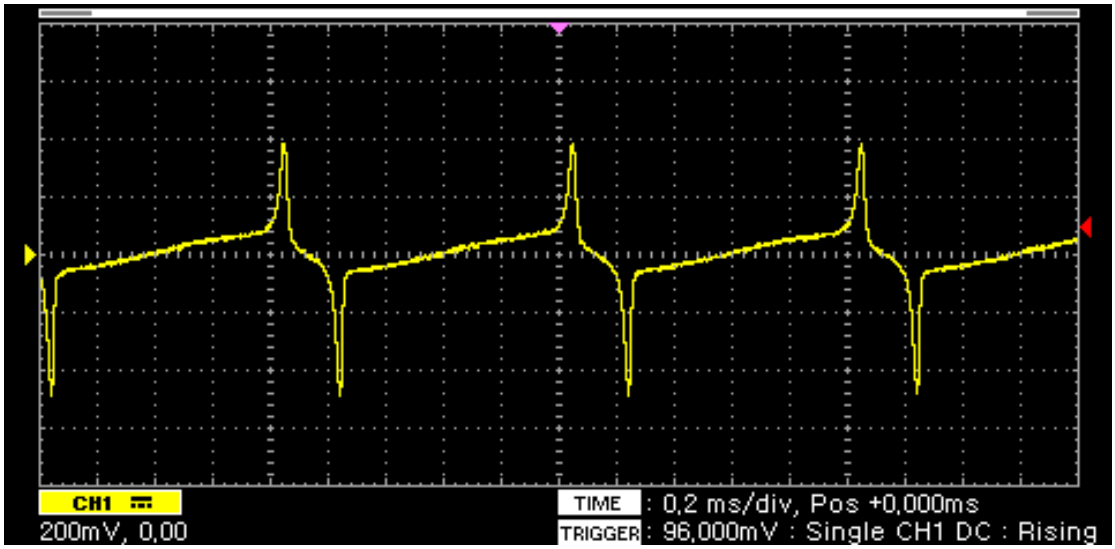
1.1 Αισθητήρας Πεδίου

1.1.1 Διεξαγωγή μετρήσεων

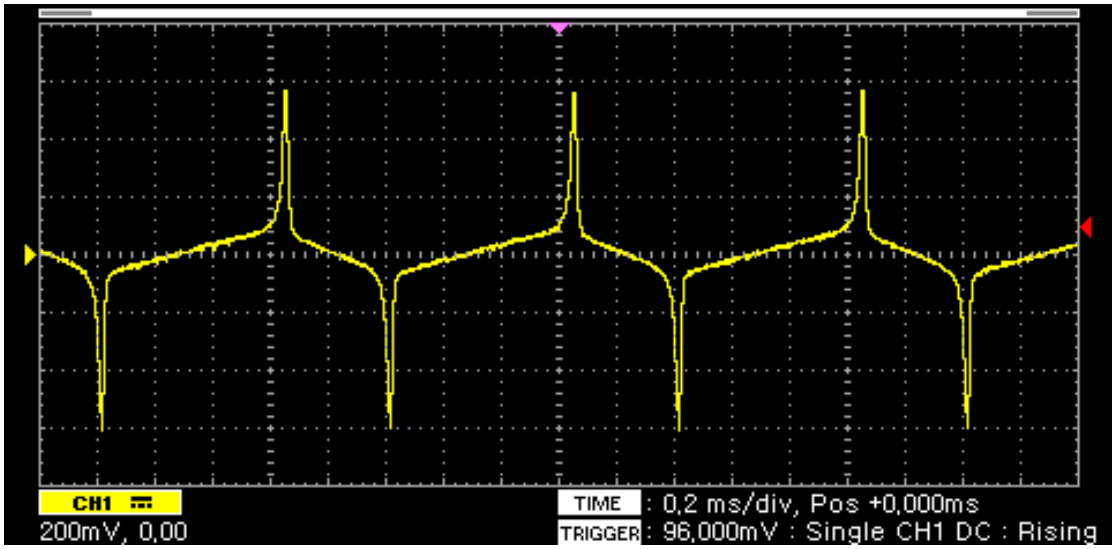
Βήμα 4:



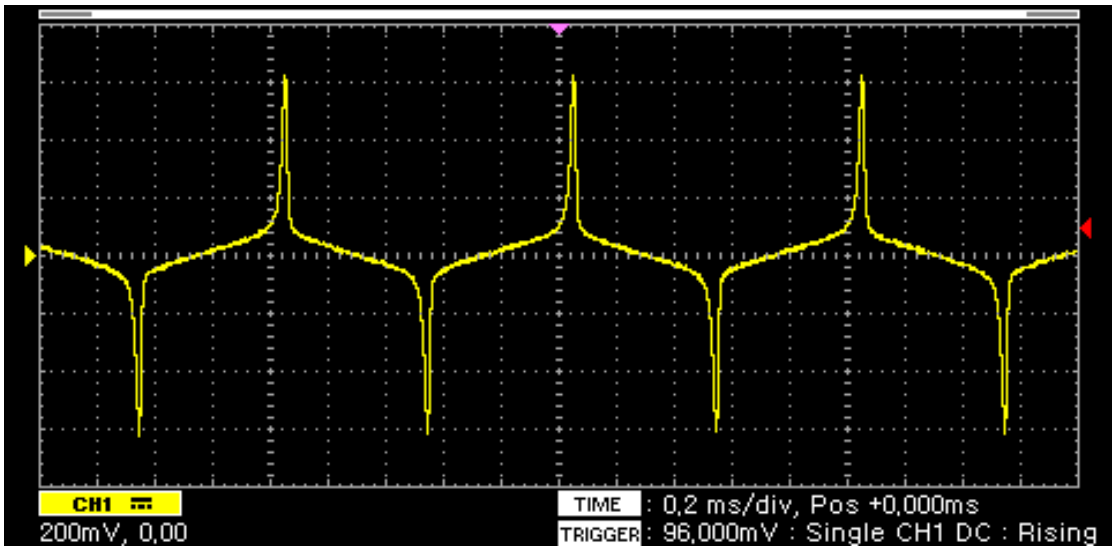
-3 V



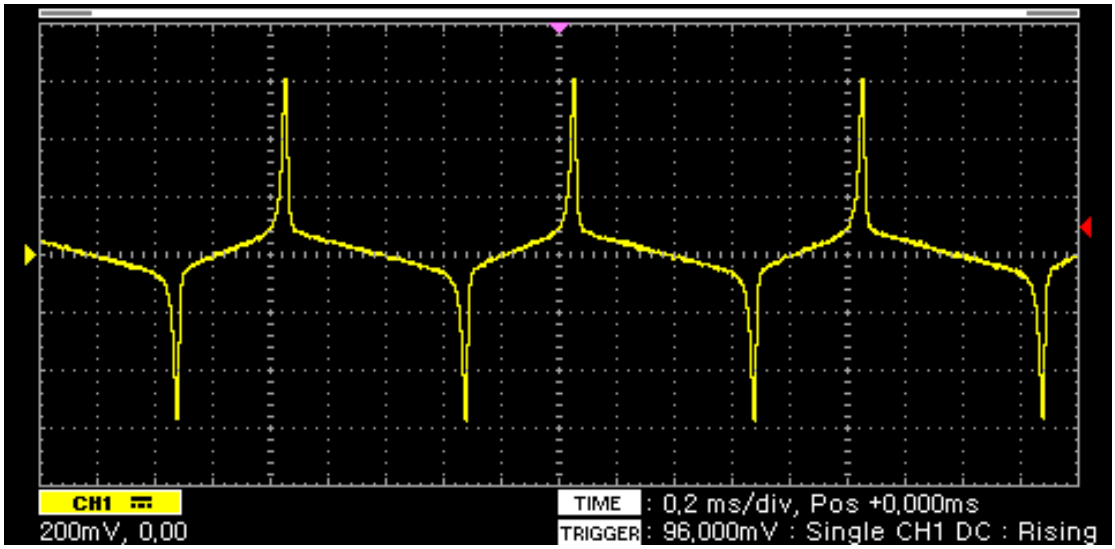
-2 V


CH1
200mV, 0,00
TIME : 0,2 ms/div, Pos +0,000ms
TRIGGER: 96,000mV : Single CH1 DC : Rising

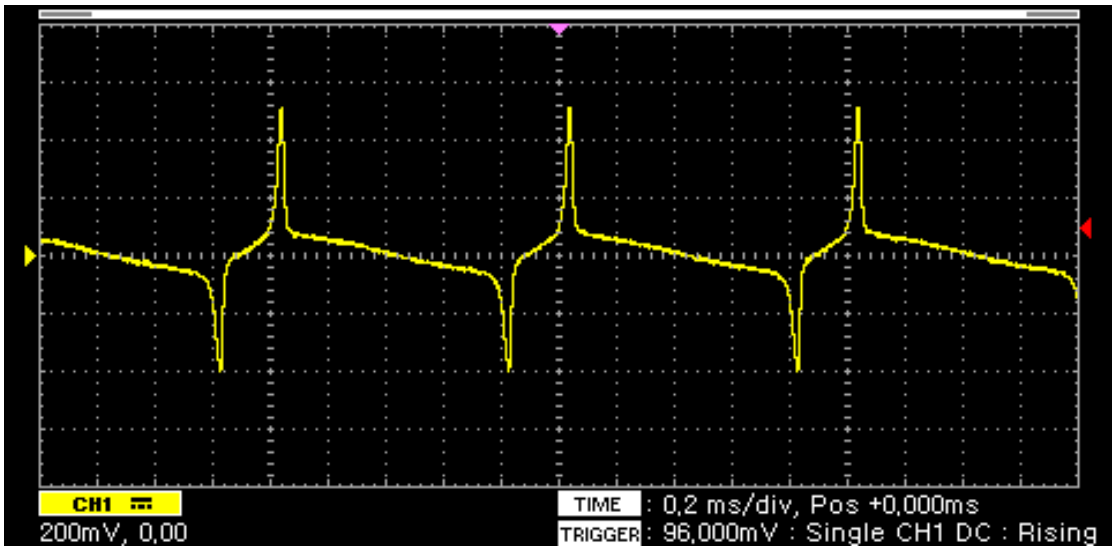
-1 V


CH1
200mV, 0,00
TIME : 0,2 ms/div, Pos +0,000ms
TRIGGER: 96,000mV : Single CH1 DC : Rising

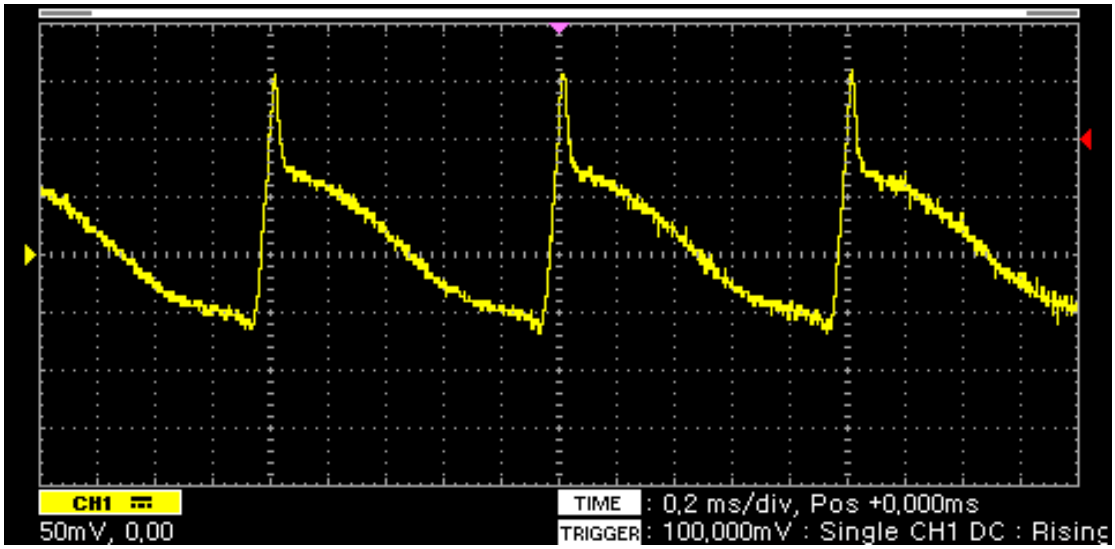
0 V


CH1
200mV, 0,00
TIME : 0,2 ms/div, Pos +0,000ms
TRIGGER: 96,000mV : Single CH1 DC : Rising

1 V

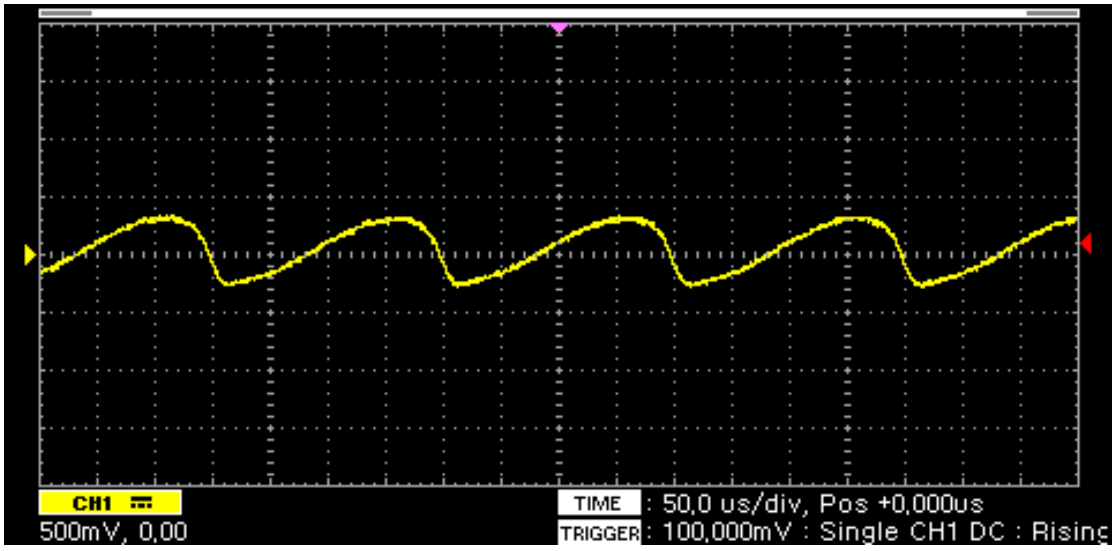

CH1
200mV, 0,00
TIME : 0,2 ms/div, Pos +0,000ms
TRIGGER: 96,000mV : Single CH1 DC : Rising

2 V

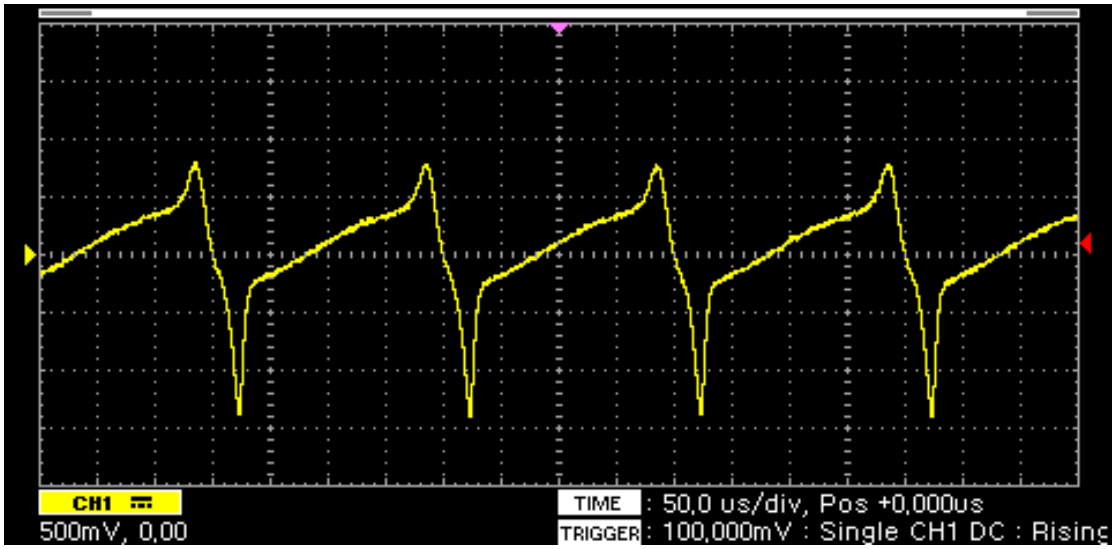

CH1
50mV, 0,00
TIME : 0,2 ms/div, Pos +0,000ms
TRIGGER: 100,000mV : Single CH1 DC : Rising

3 V

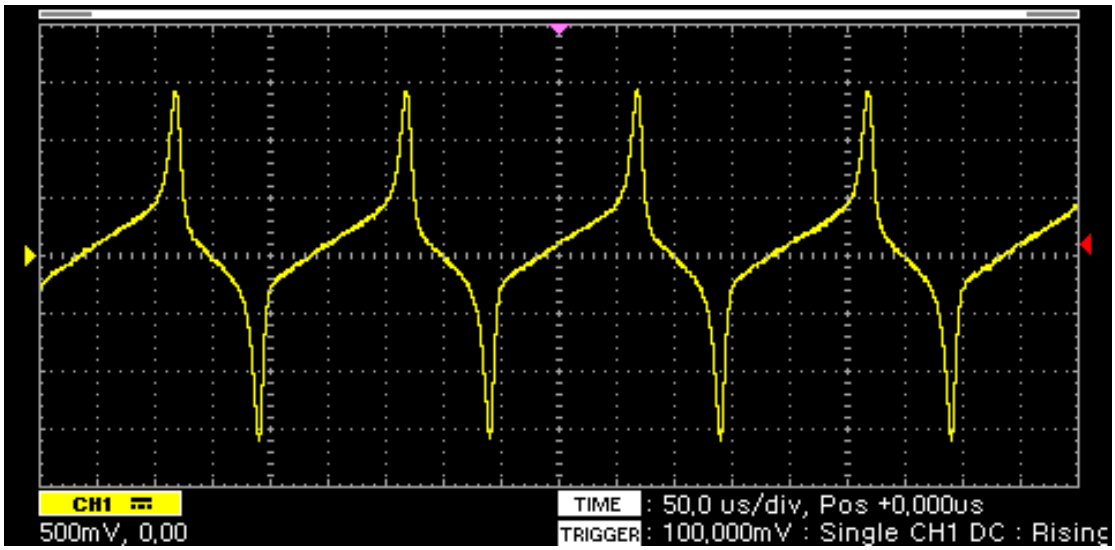
Βήμα 5:


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

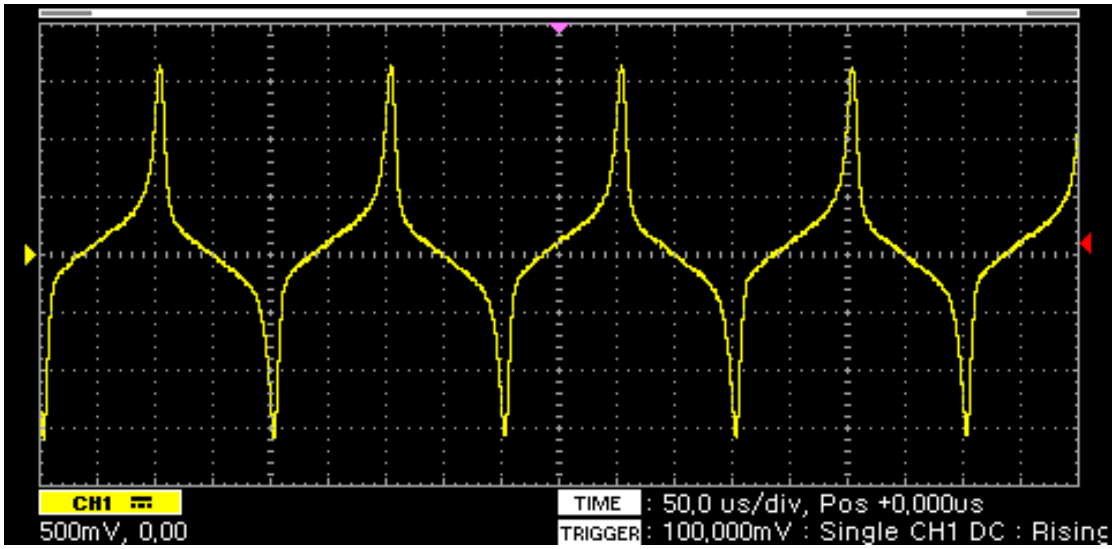
-3 V


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

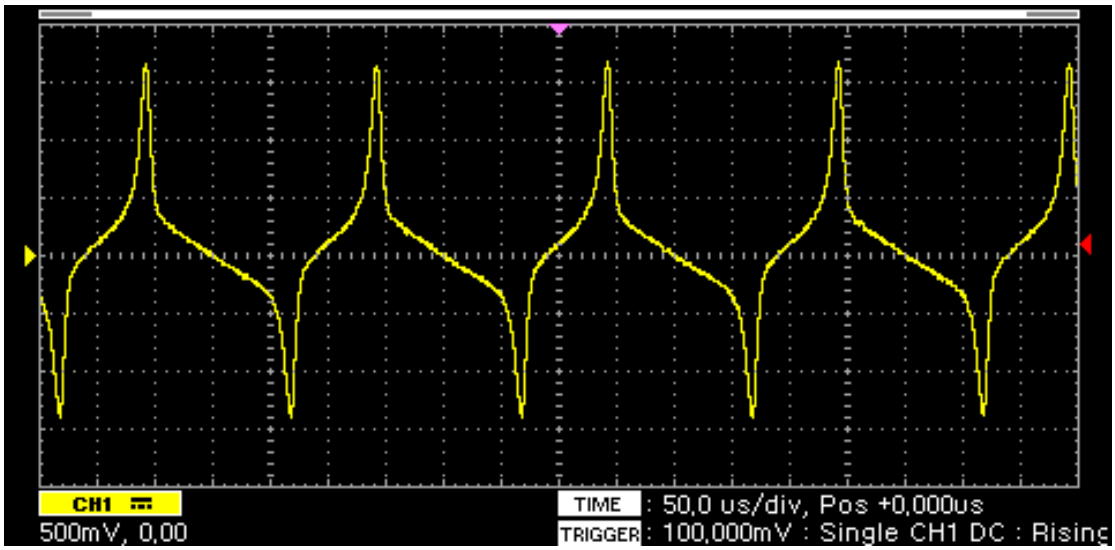
-2 V


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

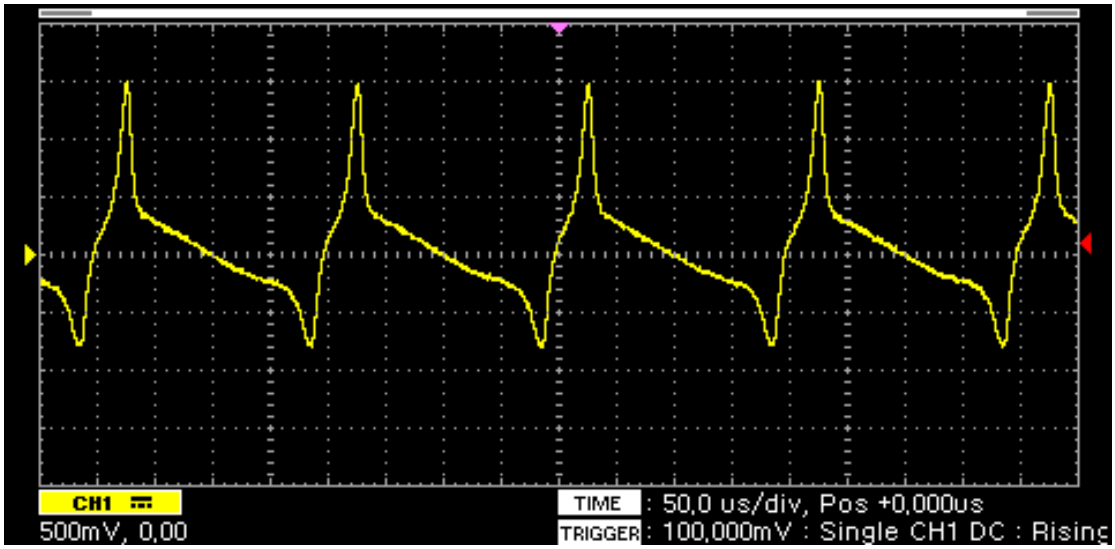
-1 V


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

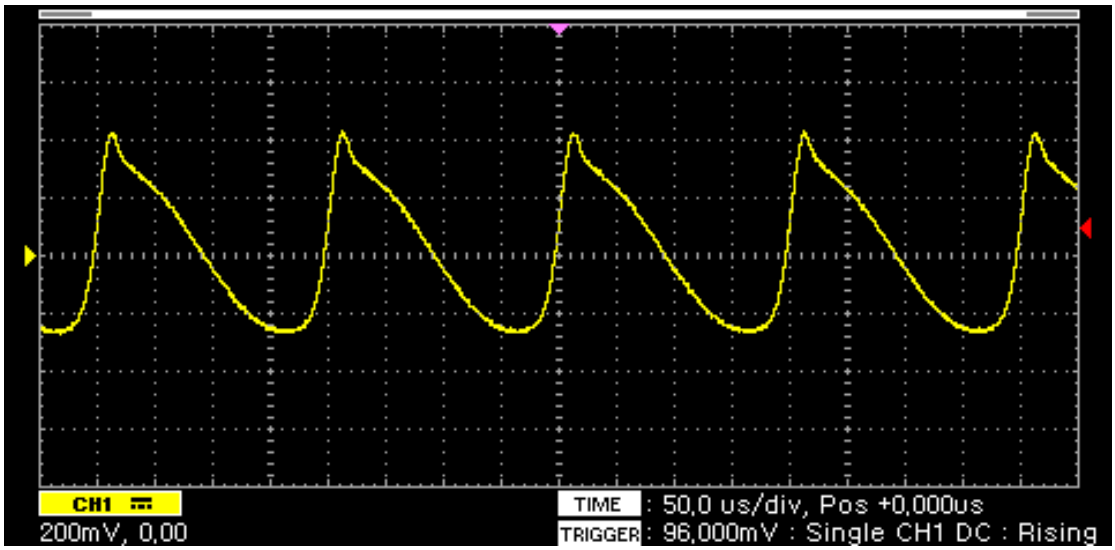
0 V


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

1 V


CH1
500mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 100,000mV : Single CH1 DC : Rising

2 V

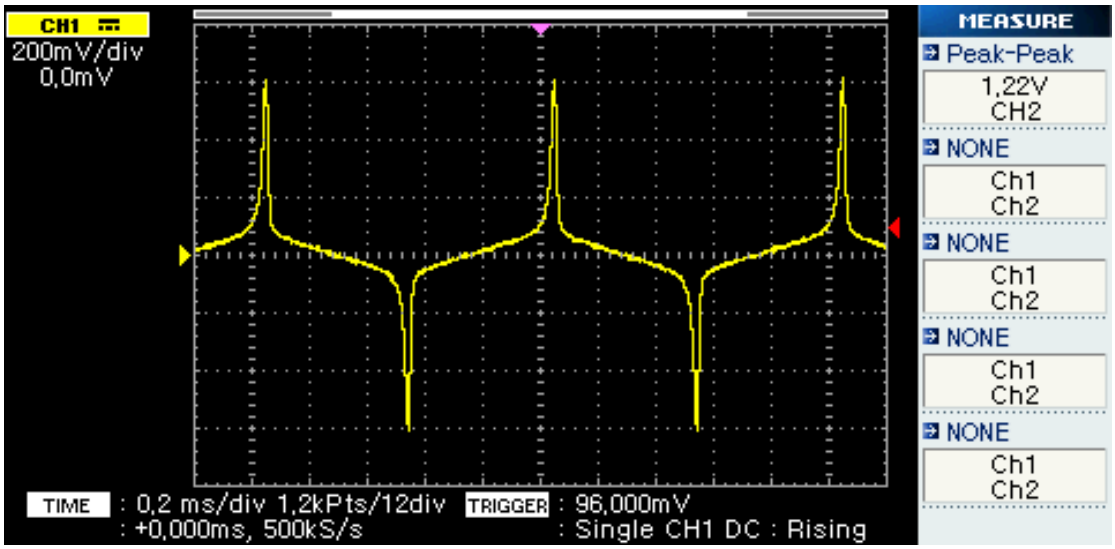

CH1
200mV, 0,00
TIME : 50,0 us/div, Pos +0,000us
TRIGGER: 96,000mV : Single CH1 DC : Rising

3 V

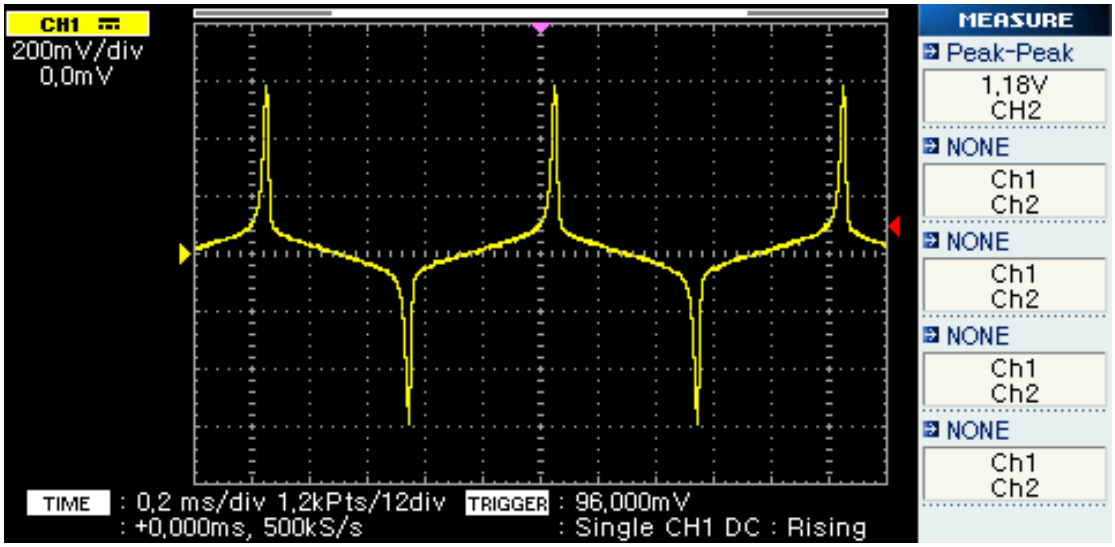
1.2 Αισθητήρας Θέσης

1.2.1 Διεξαγωγή μετρήσεων

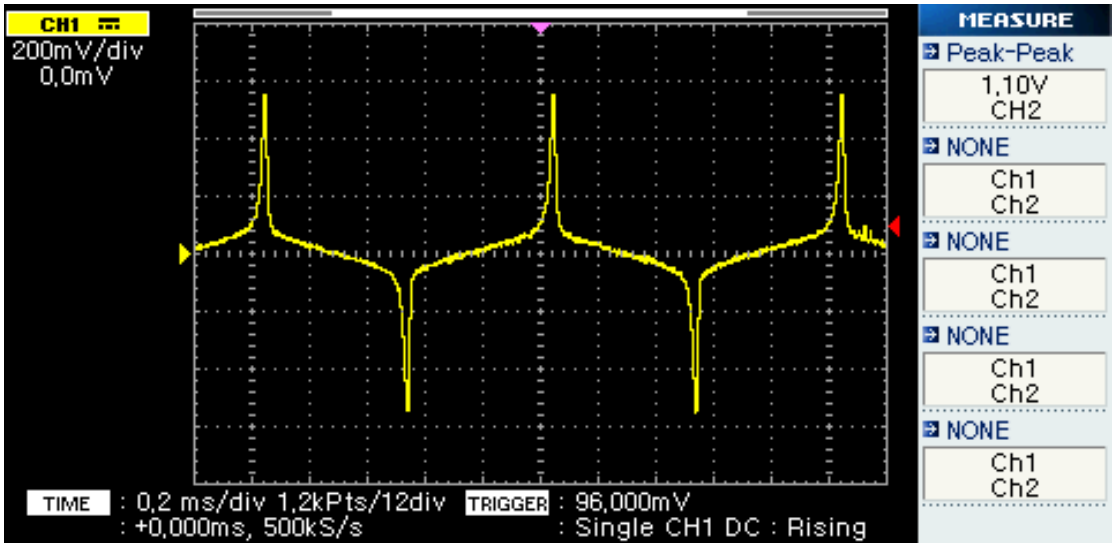
Βήμα 3:



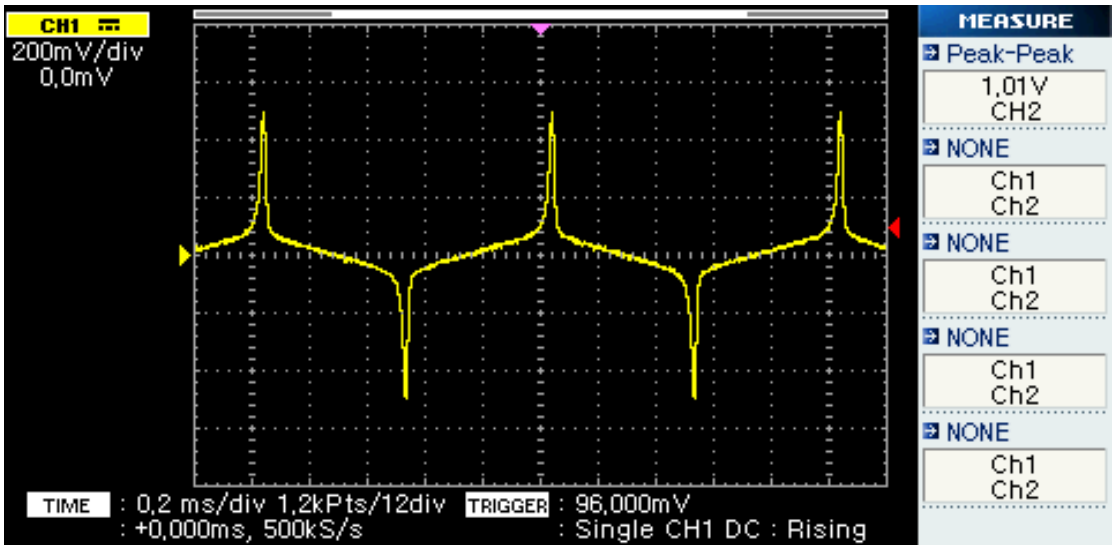
0 mm



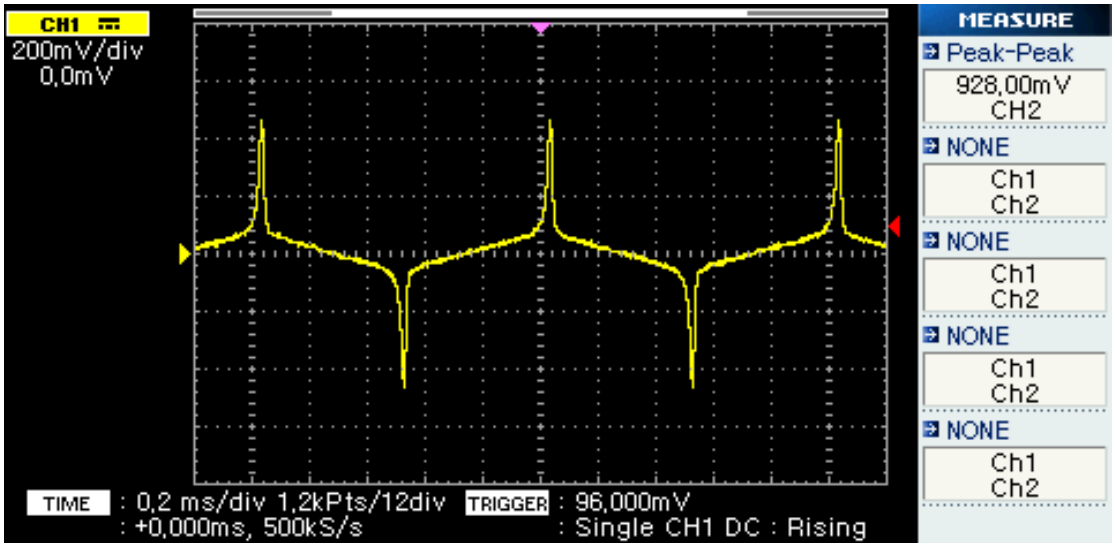
5mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
1,10V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

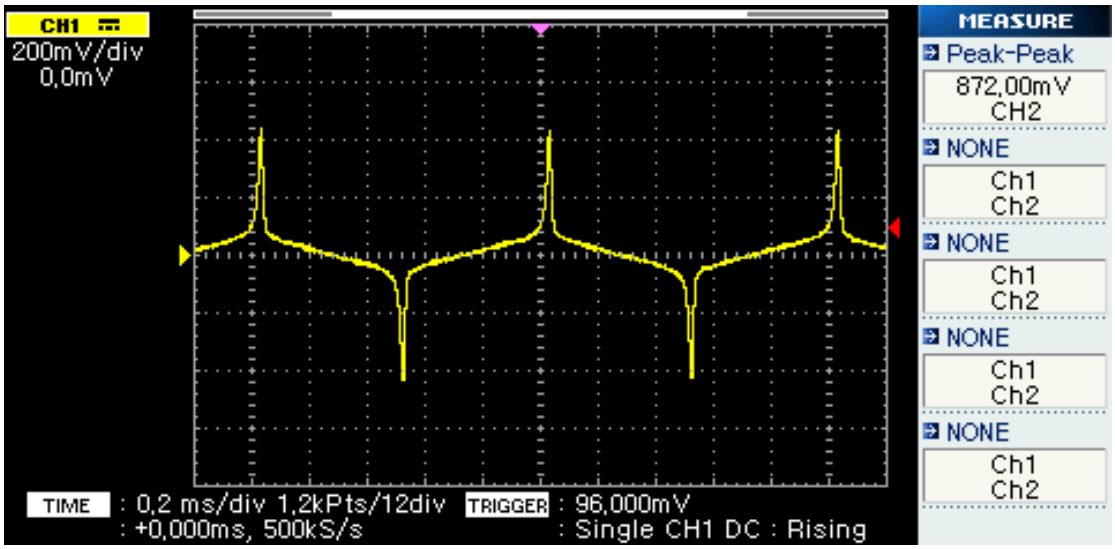
10 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
1,01V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

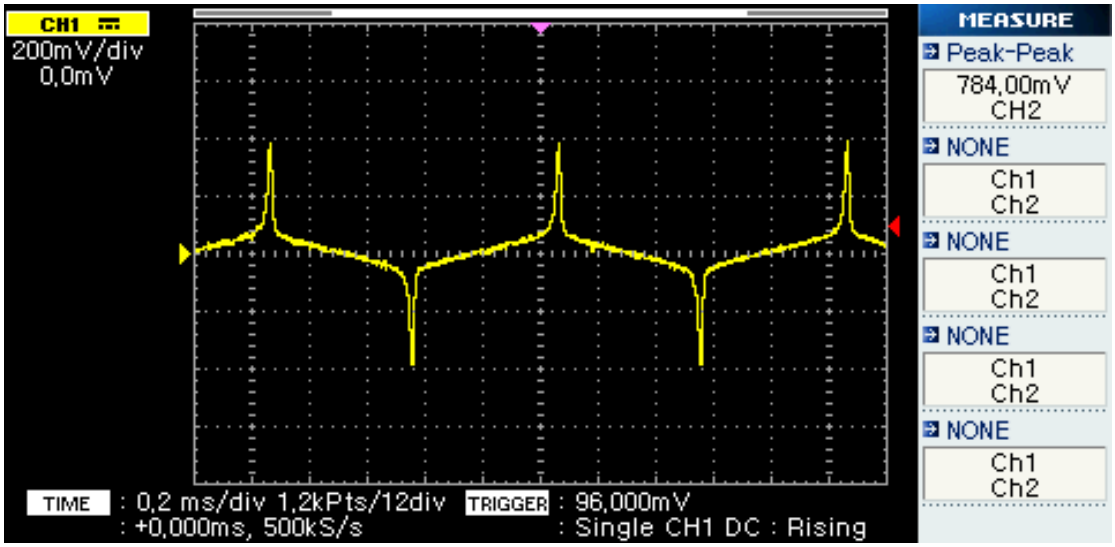
15 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
928,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

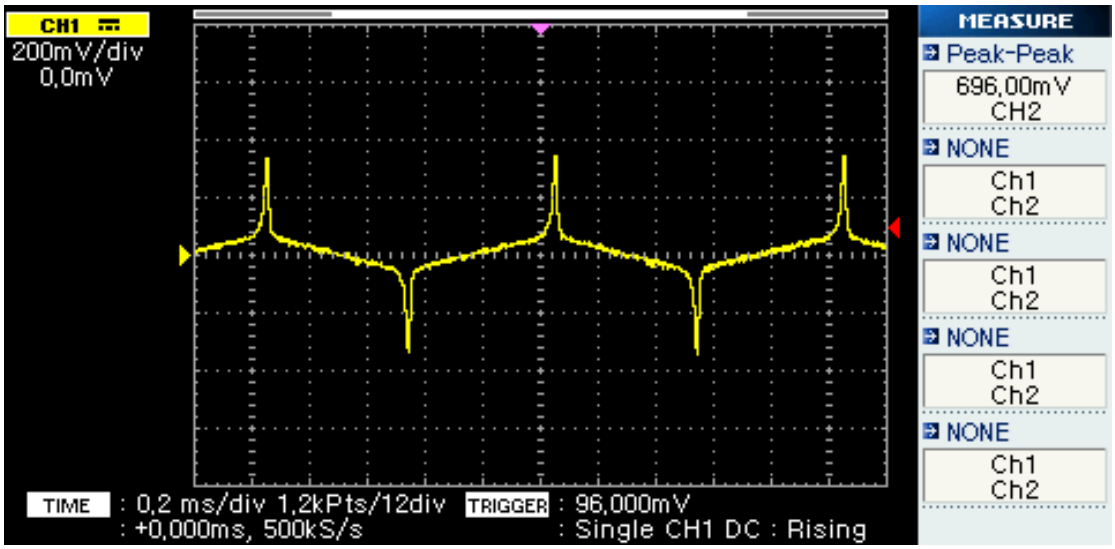
20 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
872,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

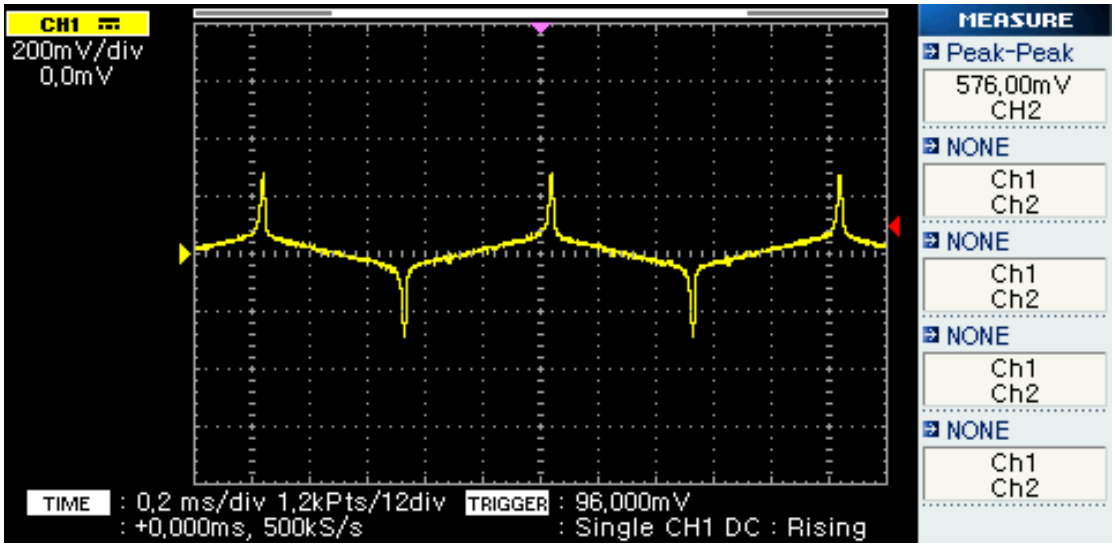
25 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
784,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

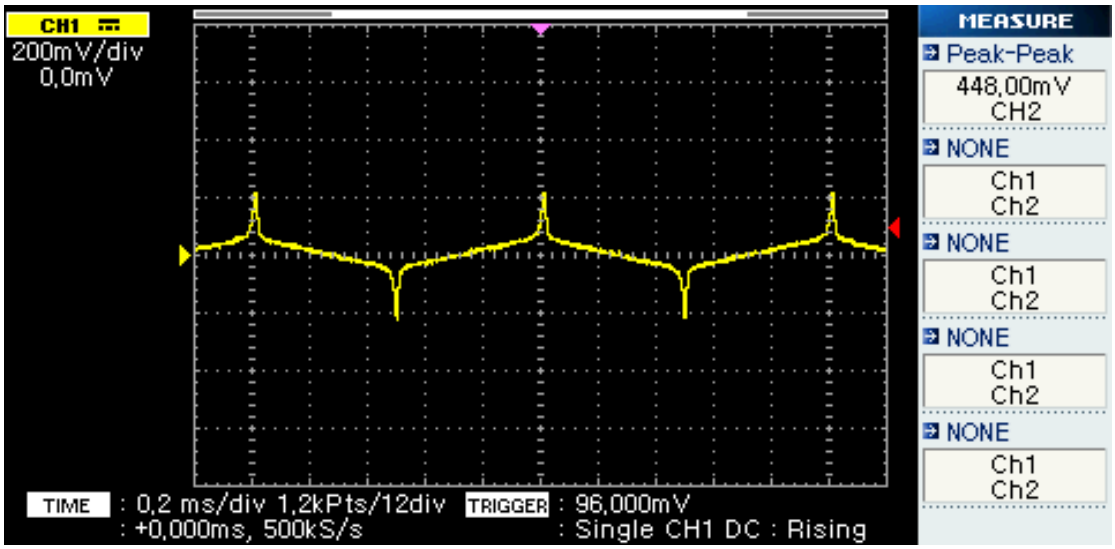
30 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
696,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

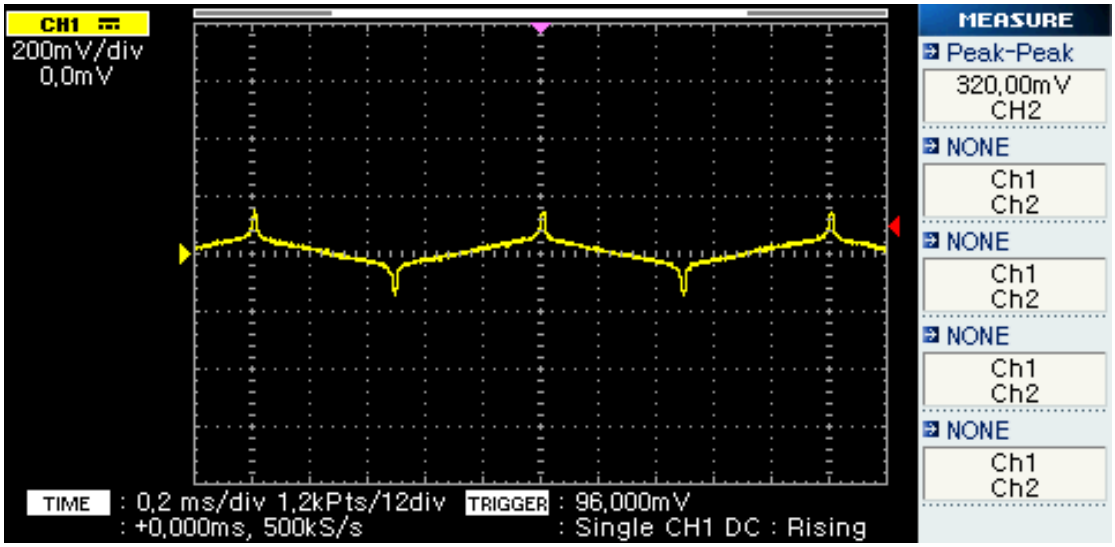
35 mm


CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
576,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

40 mm

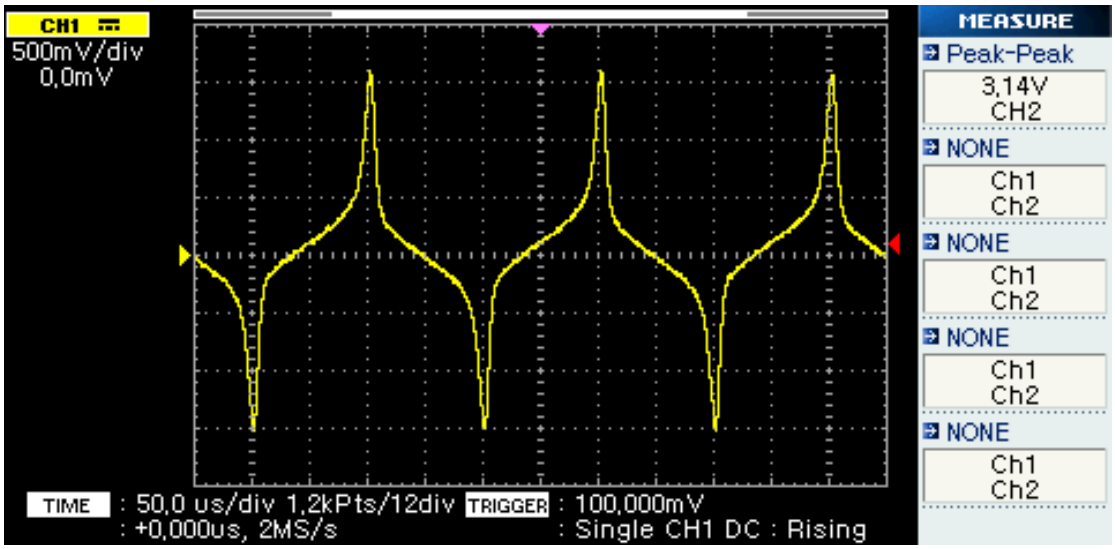

CH1
200mV/div
0,0mV
MEASURE
Peak-Peak
448,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 0,2 ms/div 1,2kPts/12div
: +0,000ms, 500kS/s
TRIGGER : 96,000mV
: Single CH1 DC : Rising

45 mm

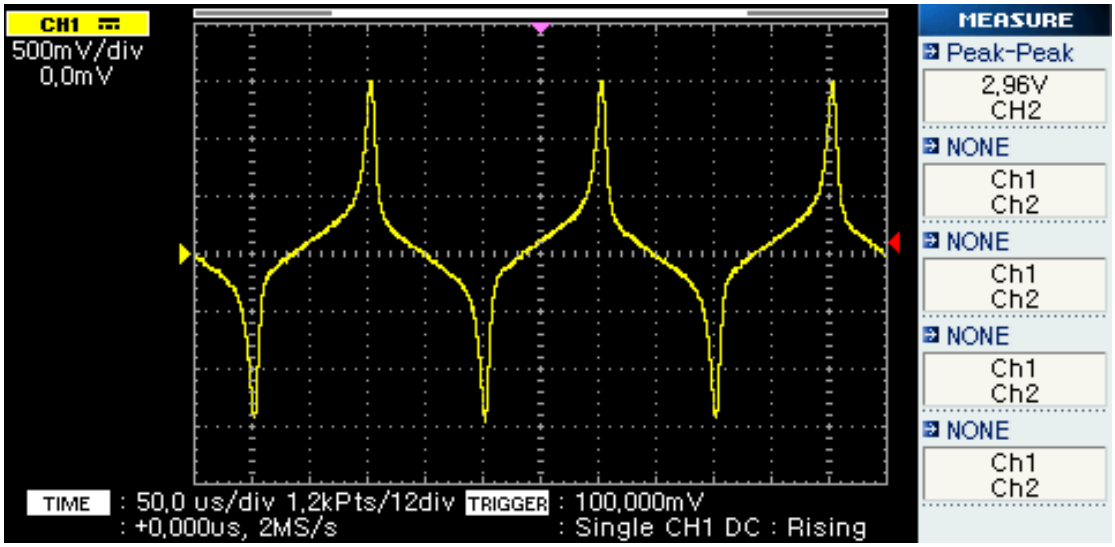


50 mm

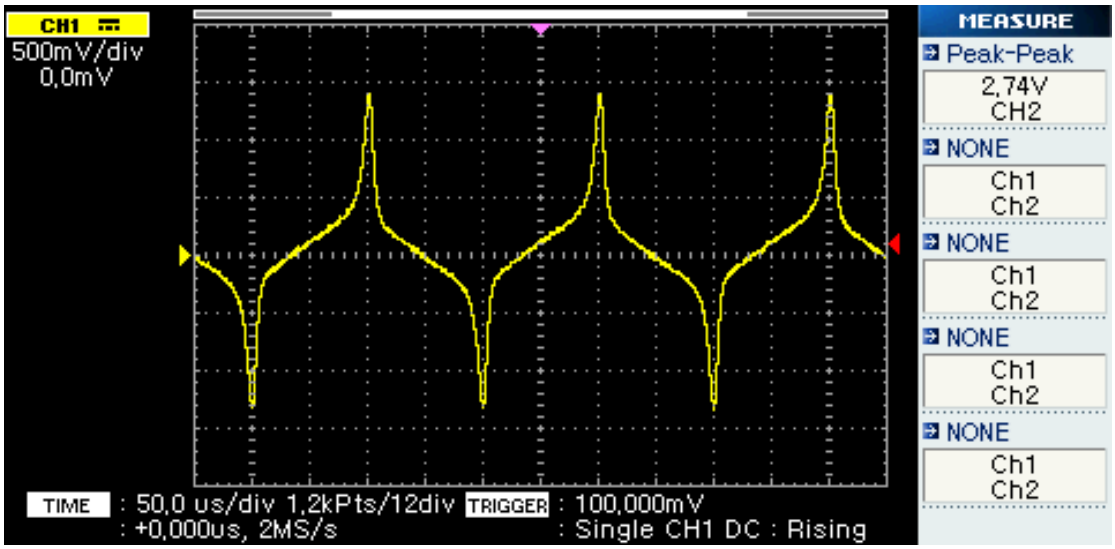
Βήμα 4:



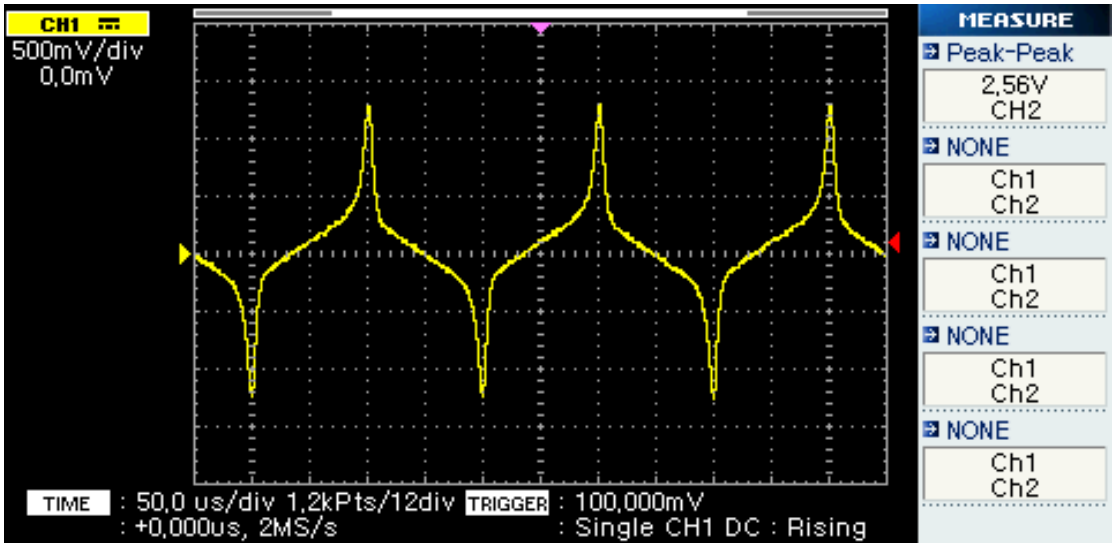
0 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
2,96V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

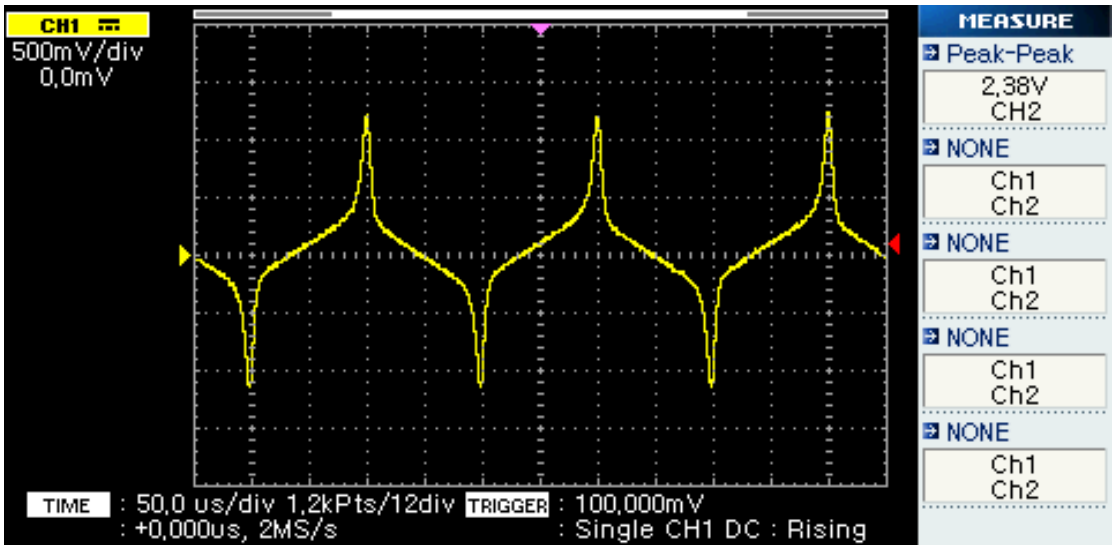
5 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
2,74V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

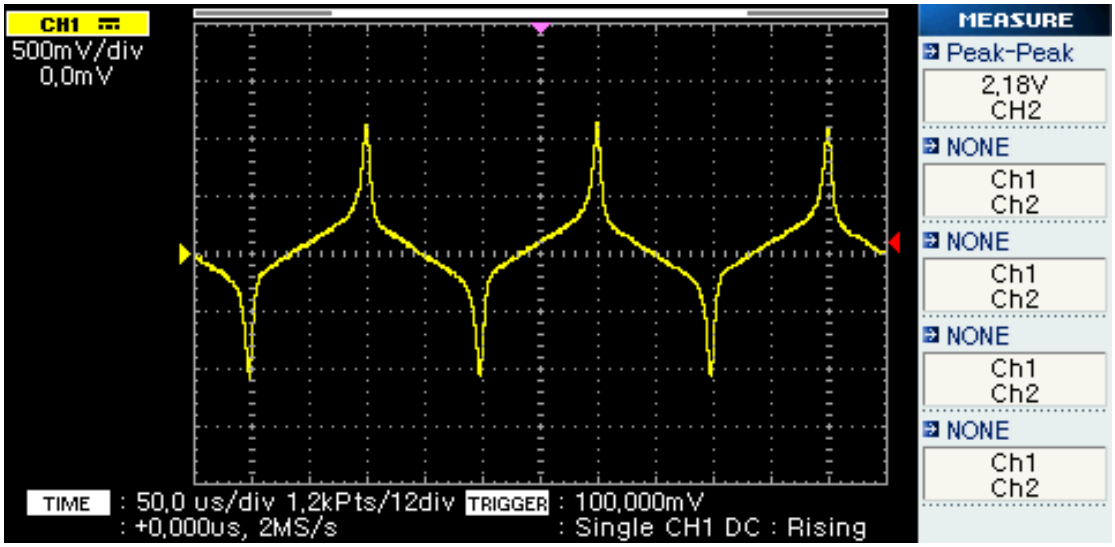
10 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
2,56V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

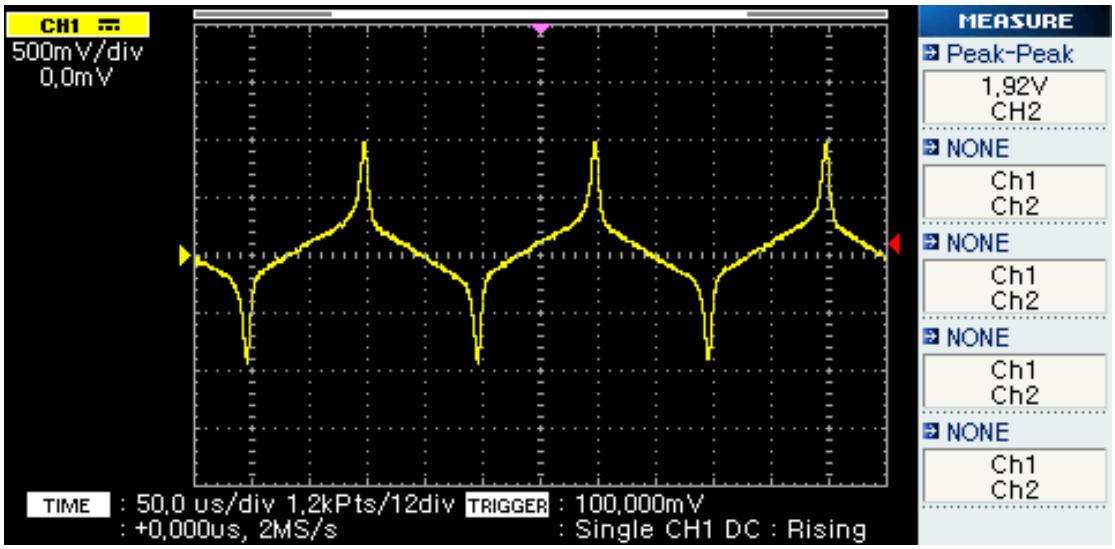
15 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
2,38V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

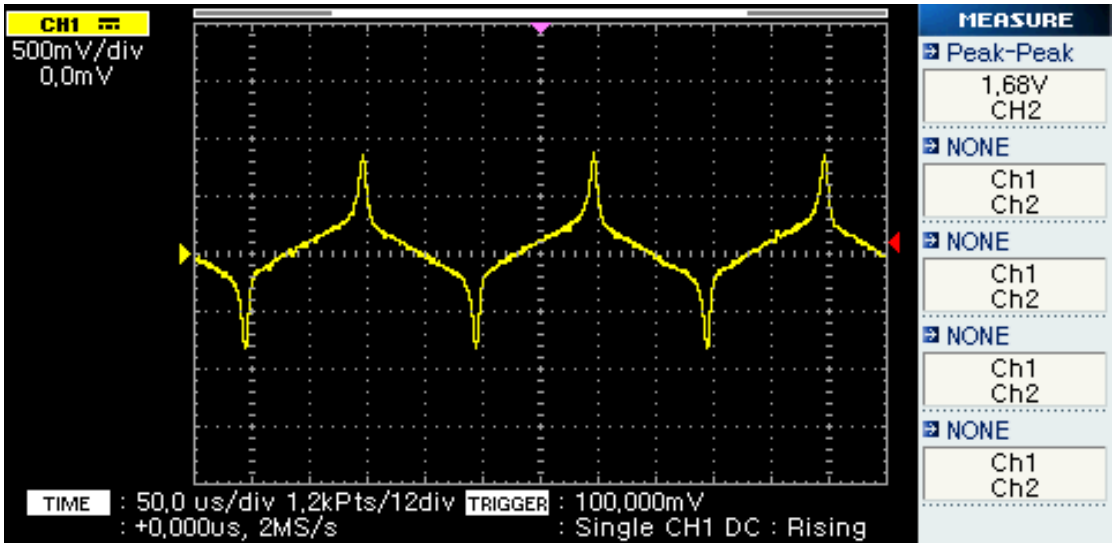
20 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
2,18V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

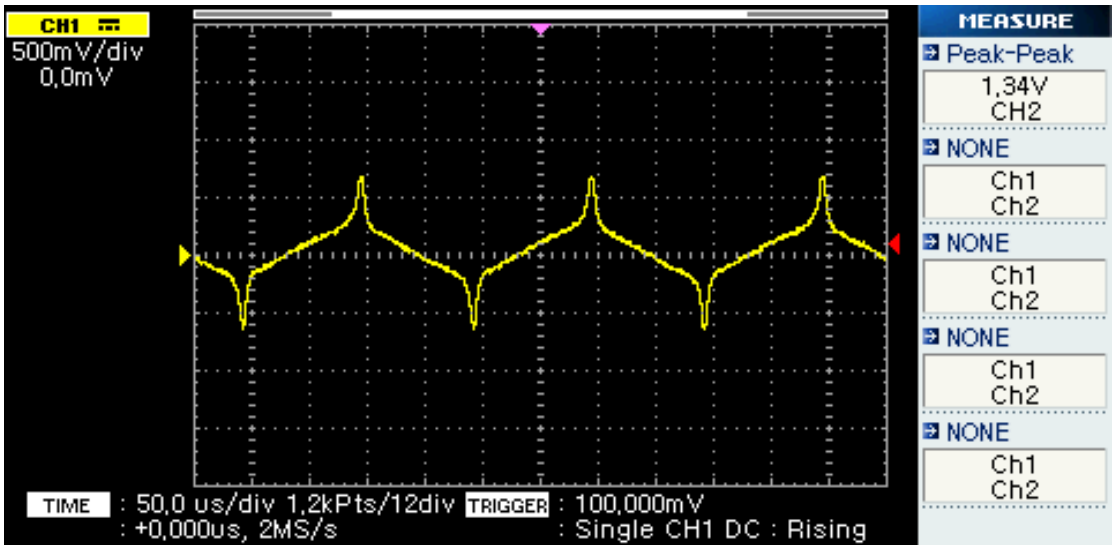
25 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
1,92V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

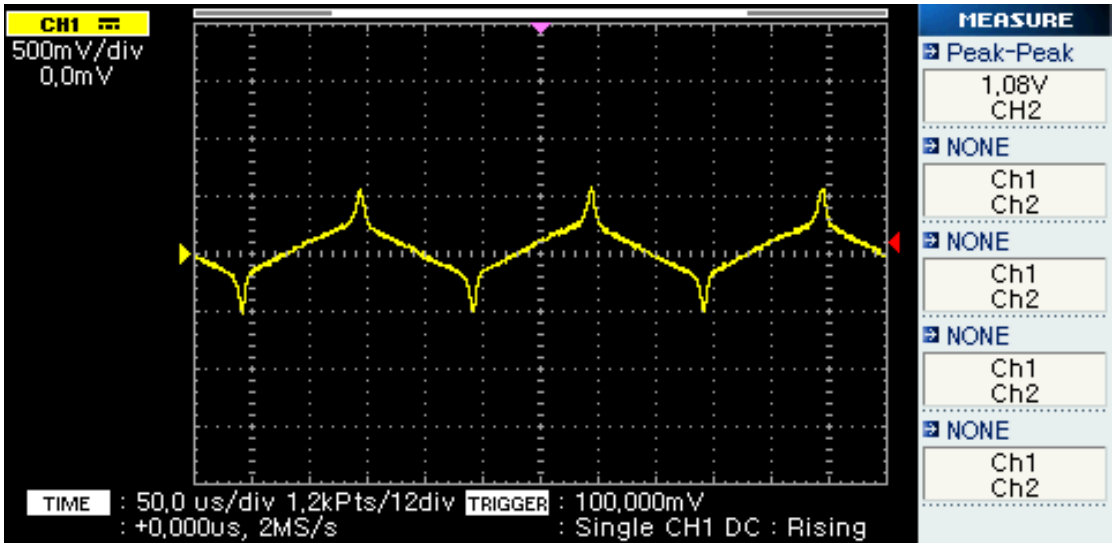
30 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
1,68V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 50,0 us/div 1,2kPts/12div TRIGGER : 100,000mV
: +0,000us, 2MS/s : Single CH1 DC : Rising

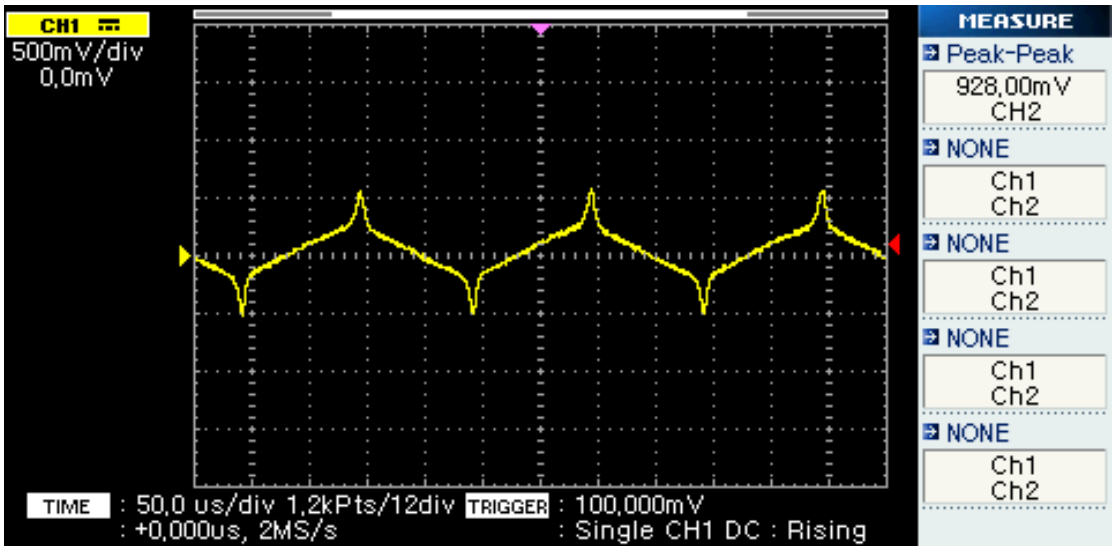
35 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
1,34V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME : 50,0 us/div 1,2kPts/12div TRIGGER : 100,000mV
: +0,000us, 2MS/s : Single CH1 DC : Rising

40 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
1,08V
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

45 mm


CH1
500mV/div
0,0mV
MEASURE
Peak-Peak
928,00mV
CH2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
NONE
Ch1
Ch2
TIME
: 50,0 us/div 1,2kPts/12div
: +0,000us, 2MS/s
TRIGGER
: 100,000mV
: Single CH1 DC : Rising

50 mm